

Experiment 6: Cloud Management & User Management

Objective: To understand administrative features of cloud platform.

Theory: Cloud management involves monitoring, provisioning, and managing user roles and permissions.

Username and Password

The credentials represent an identity in the Cloud Identity and Access Management (Cloud IAM) service. This identity has access permissions (a role or roles) that allow you to work with Google Cloud resources in the project you've been allocated

Project ID

A [Google Cloud project](#) is an organizing entity for your Google Cloud resources. It often contains resources and services; for example, it may hold a pool of virtual machines, a set of databases, and a network that connects them together. Projects also contain settings and permissions, which specify security rules and who has access to what resources.

A *Project ID* is a unique identifier that is used to link Google Cloud resources and APIs to your specific project.

Roles and permissions

In addition to cloud computing services, Google Cloud also contains a collection of permissions and roles that define who has access to what resources. You can use the [Cloud Identity and Access Management \(Cloud IAM\)](#) service to inspect and modify roles and permissions.

Basic roles set project-level permissions and, unless otherwise specified, control access and management to all Google Cloud services. The following table pulls definitions from the [roles documentation](#), which gives a brief overview of `viewer`, `editor`, and `owner` role permissions.

Role Name	Permissions
roles/viewer	Permissions for read-only actions that do not affect state, such as viewing (but not modifying) existing resources or data.
roles/editor	All viewer permissions, plus permissions for actions that modify state, such as changing existing resources.
roles/owner	All editor permissions and permissions for the following actions: manage roles and permissions for a project and all resources within the project; set up billing for a project.

As an editor, you can create, modify, and delete Google Cloud resources. However, you can't add or delete members from Google Cloud projects.

Procedure:

1. Access cloud dashboard (Google Cloud Platform):
https://www.cloudskillsboost.google/focuses/2794?catalog_rank=%7B%22rank%22%3A4%2C%22num_filters%22%3A0%2C%22has_search%22%3Atrue%7D&parent=catalog&search_id=53245987
2. Access the Cloud console with specific credentials to explore the lab platform and identify key features of a lab environment.
3. View various Google Cloud projects and identify common misconceptions about them.
4. Use the Google Cloud console navigation menu to identify types of Google Cloud services.
5. Manage basic roles and use the Cloud IAM service to inspect actions available to specific users.
6. Explore the API library and examine its chief features.

Viva Questions:

- What is cloud management?
- What is IAM ?
- Why is user role management important?